What Separates Red from White, and What Lies Beneath?

A Multivariate Analysis (MANOVA, PCA, and EFA) of the Physicochemical Characteristics of Portuguese Vinho Verde Wines

APM1220 – Applied Multivariate Data Analysis · Summative Assessment 1

1. Introduction

Wine quality is judged by trained tasters, but the sensory impression rests on measurable chemistry: acidity shapes freshness, residual sugar and alcohol shape body and sweetness, and sulfur dioxide protects the wine against oxidation and microbial spoilage. Because these quantities are measured together and are physically and biologically linked, they are naturally studied with multivariate methods rather than one variable at a time. Vinho Verde is a Portuguese wine from the Minho region, produced in both red and white styles, and its physicochemical profile has been used to model sensory quality (Cortez et al., 2009).

The purpose of this study is to characterize the physicochemical structure of Vinho Verde wines using three complementary techniques applied to the same data: multivariate analysis of variance (MANOVA) to test whether red and white wines differ in their joint chemical profile, principal component analysis (PCA) to summarize the 11 measurements with fewer dimensions, and exploratory factor analysis (EFA) to examine the latent dimensions that may generate the observed correlations. Sensory quality is deliberately *not* modelled here; the analyses concern chemistry only.

2. Dataset

The data are the Wine Quality Dataset from the UCI Machine Learning Repository (https://archive.ics.uci.edu/dataset/186/wine+quality), released under a CC BY 4.0 licence (Cortez et al., 2009). The two files were combined into a single analytic dataset of **N = 6,497** wines: **1,599 red** (24.6%) and **4,898 white** (75.4%). Each record contains 11 continuous physicochemical variables (fixed acidity, volatile acidity, citric acid, residual sugar, chlorides, free sulfur dioxide, total sulfur dioxide, density, pH, sulphates, alcohol), a sensory quality score (integer, 0–10, median of at least three assessors), and a wine-color label (Red/White) that was added when the files were merged, giving 13 variables in all. Quality is retained in the file but excluded from MANOVA, PCA, and EFA, as the assessment requires. The group imbalance (about 1 red wine for every 3 white wines) is addressed in the MANOVA assumption checks.

3. Research Questions and Hypotheses

- **RQ1 (MANOVA).** Do red and white wines differ significantly in their overall physicochemical profile when several characteristics are considered simultaneously? H_0: the mean vector of the selected physicochemical variables is the same for red and white wines ($\boldsymbol{\mu}_{red} = \boldsymbol{\mu}_{white}$). H_1: the mean vectors differ.
- **RQ2 (PCA).** Can the 11 physicochemical measurements be summarized by a smaller number of principal components that retain a substantial proportion of the total variability? H_0: the variables are uncorrelated (correlation matrix = identity), so no components can be extracted usefully. H_1: the variables are sufficiently correlated for a low-dimensional summary (tested with Bartlett's test and judged by retained variance).
- **RQ3 (EFA).** What underlying latent dimensions explain the relationships among the physicochemical characteristics? H_0: no common factors are needed (correlation matrix = identity). H_1: a small number of common factors accounts for the inter-variable correlations.

4. Methodology

Software. All analyses were run in Python 3.12 using pandas, NumPy, SciPy, statsmodels (MANOVA), pingouin (Henze–Zirkler test), scikit-learn-compatible factor_analyzer (KMO, Bartlett, EFA), matplotlib, and seaborn. A fixed random seed (2026) makes the simulation-based parallel analyses reproducible. The full script is *code/SA1_Wine_Analysis.py*.

Data preparation and exploration. The two CSV files (semicolon-delimited) were imported, labelled by color, and concatenated. I checked structure, missing values, and exact duplicate rows; computed descriptive statistics (mean, SD, range, skewness, excess kurtosis, coefficient of variation); screened univariate outliers (Tukey's 1.5×IQR rule and $|z| > 3.29$) and multivariate outliers (Mahalanobis D^2 against a χ^2 criterion at $p < .001$); and produced density plots, boxplots by color, a scatterplot matrix, and a correlation heatmap.

MANOVA. Color was the grouping factor. Dependent variables were selected from the exploratory results (Section 6.2). Assumptions were examined through design considerations and duplicate checks (independence), Henze–Zirkler tests and χ^2 Q–Q plots of Mahalanobis distances (multivariate normality), Mahalanobis screening (outliers), Box's *M* and Levene's tests (homogeneity of covariance), and the correlation matrix and its determinant (relationships among DVs). The omnibus test was Pillai's trace, chosen for its robustness to violations; Wilks' Λ, Hotelling–Lawley, and Roy's root are also reported. Follow-up analyses were univariate ANOVAs with a Bonferroni correction ($\alpha = .05/7 = .0071$), Welch tests as a robustness check, η^2 and Cohen's *d*, and the discriminant function's standardized coefficients and structure correlations. Sensitivity analyses repeated the MANOVA without duplicate rows and without multivariate outliers.

PCA. The 11 variables were z-standardized (so that the analysis operates on the correlation matrix and no variable dominates through its measurement scale). Factorability was checked with Bartlett's test and KMO. The number of components was decided jointly from eigenvalues, the scree plot, cumulative variance, and Horn's parallel analysis (1,000 random normal datasets, 95th percentile) rather than the Kaiser rule alone.

EFA. Factorability was assessed with the correlation matrix, KMO, and Bartlett's test. The number of factors was judged by the scree plot, parallel analysis, eigenvalues, and interpretability. I used maximum-likelihood (ML) extraction, which provides fit statistics, with **oblimin** (oblique) rotation because chemical dimensions need not be uncorrelated. Robustness was checked with minimum-residual (minres) extraction and with varimax rotation, and the structure was re-examined after removing the red/white mean difference (pooled within-color data).

5. Results

5.1 Exploratory Data Analysis

The combined file has 6,497 observations and 13 variables (11 continuous physicochemical variables, quality, and color). There were **no missing values**. There were **1,177 exact duplicate rows** (18.1%; 937 white, 240 red). Because the data lack a sample identifier, these cannot be distinguished from different wines that happened to have identical readings, so they were retained in the main analyses (the assessment specifies the full 6,497 observations), and their influence was examined in a sensitivity analysis (Section 6.2).

Table 1. Descriptive statistics for the 11 physicochemical variables (N = 6,497)

Variable	M	SD	Min	Max	Skew	Ex. kurt.	CV (%)	Outliers (IQR)	Outliers (|z|>3.29)
fixed acidity	7.22	1.30	3.80	15.90	1.72	5.06	18.0	357	111
volatile acidity	0.34	0.16	0.08	1.58	1.50	2.83	48.5	377	62
citric acid	0.32	0.15	0.00	1.66	0.47	2.40	45.6	509	19
residual sugar	5.44	4.76	0.60	65.80	1.44	4.36	87.4	118	9
chlorides	0.0560	0.0350	0.0090	0.6110	5.40	50.90	62.5	286	88
free sulfur dioxide	30.53	17.75	1.00	289.00	1.22	7.91	58.1	62	27
total sulfur dioxide	115.74	56.52	6.00	440.00	-0.00	-0.37	48.8	10	6
density	0.9947	0.0030	0.9871	1.0390	0.50	6.61	0.3	3	3
pH	3.22	0.16	2.72	4.01	0.39	0.37	5.0	73	19
sulphates	0.53	0.15	0.22	2.00	1.80	8.65	28.0	191	56

alcohol	10.49	1.19	8.00	14.90	0.57	-0.53	11.4	3	1

Variability. On a unit-free basis, residual sugar (CV = 87%) is the most variable variable, followed by chlorides (63%) and free sulfur dioxide (58%); volatile acidity, total sulfur dioxide, and citric acid cluster near 46–49%. Density (CV = 0.3%) and pH (5%) are tightly constrained. In absolute terms, total sulfur dioxide (SD = 56.5) and free sulfur dioxide (SD = 17.8) dominate because of their measurement scale, which is exactly why standardization is required before PCA and EFA.

Distributions. Several variables are strongly right-skewed with heavy tails: chlorides (skew 5.4, excess kurtosis 50.9), sulphates (1.8, 8.7), fixed acidity (1.7, 5.1), volatile acidity (1.5), residual sugar (1.4, 4.4), and free sulfur dioxide (1.2, 7.9). Total sulfur dioxide, pH, citric acid, and alcohol are comparatively symmetric. Part of the skew and the multimodality visible in Figure 1 comes from mixing two wine types.

Figure 1. Density of physicochemical variables by wine color

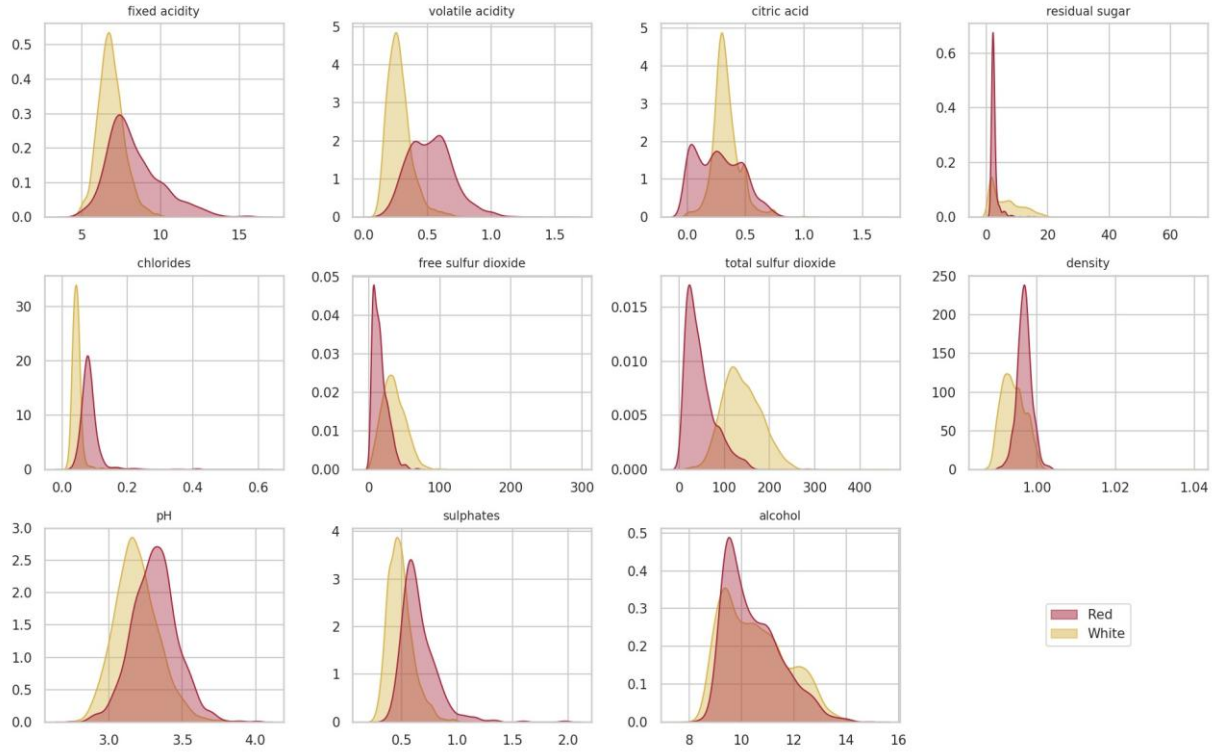


Figure 1 shows the group-specific densities: the red and white distributions overlap little for total sulfur dioxide, volatile acidity, chlorides, and fixed acidity, but almost completely for alcohol.

Figure 2. Boxplots of physicochemical variables by wine color

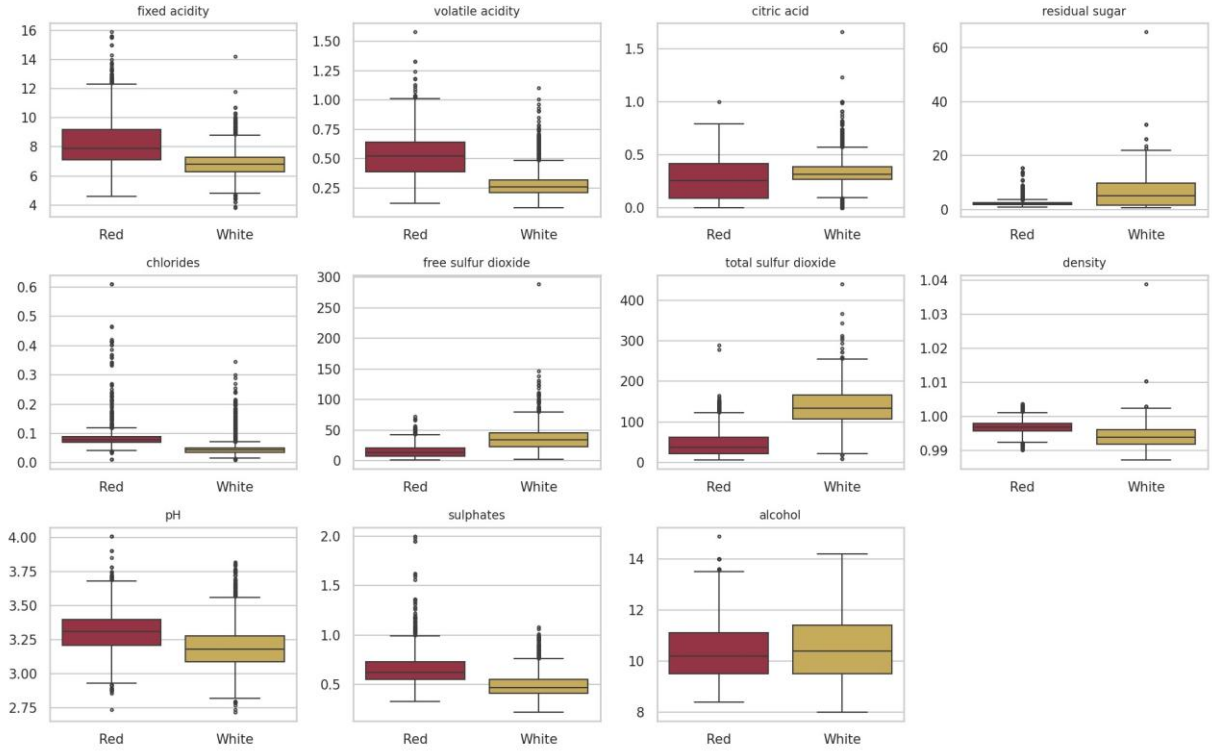


Figure 2 shows that red wines have higher fixed and volatile acidity, chlorides, sulphates, density, and pH, and lower residual sugar, free and total sulfur dioxide; alcohol medians are almost identical.

Red versus white. Red wines have clearly higher fixed acidity (8.32 vs 6.86), volatile acidity (0.53 vs 0.28), chlorides (0.087 vs 0.046), and sulphates (0.66 vs 0.49), and slightly higher density (0.9967 vs 0.9940). Red wines also have a somewhat higher pH (3.31 vs 3.19) despite their higher fixed acidity, which suggests a different acid composition. White wines have far more total sulfur dioxide (138.4 vs 46.5 mg/L) and residual sugar (6.39 vs 2.54 g/L, with a much longer right tail). Alcohol is nearly the same (10.42 vs 10.51).

Figure 3. Pearson correlation matrix (combined data, n = 6,497)

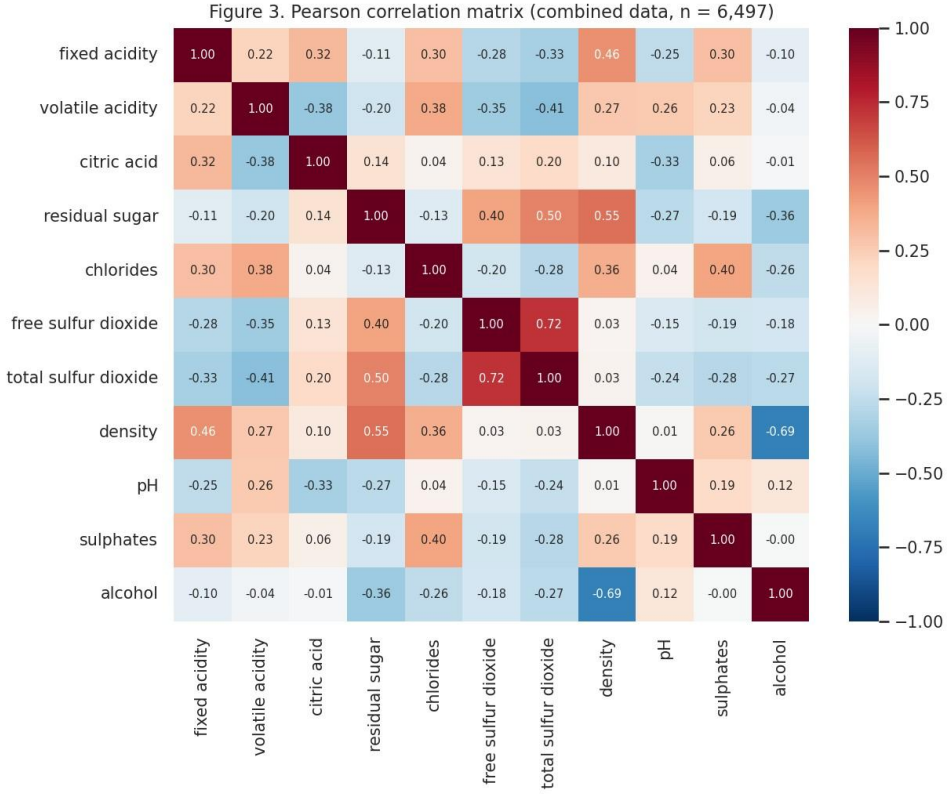
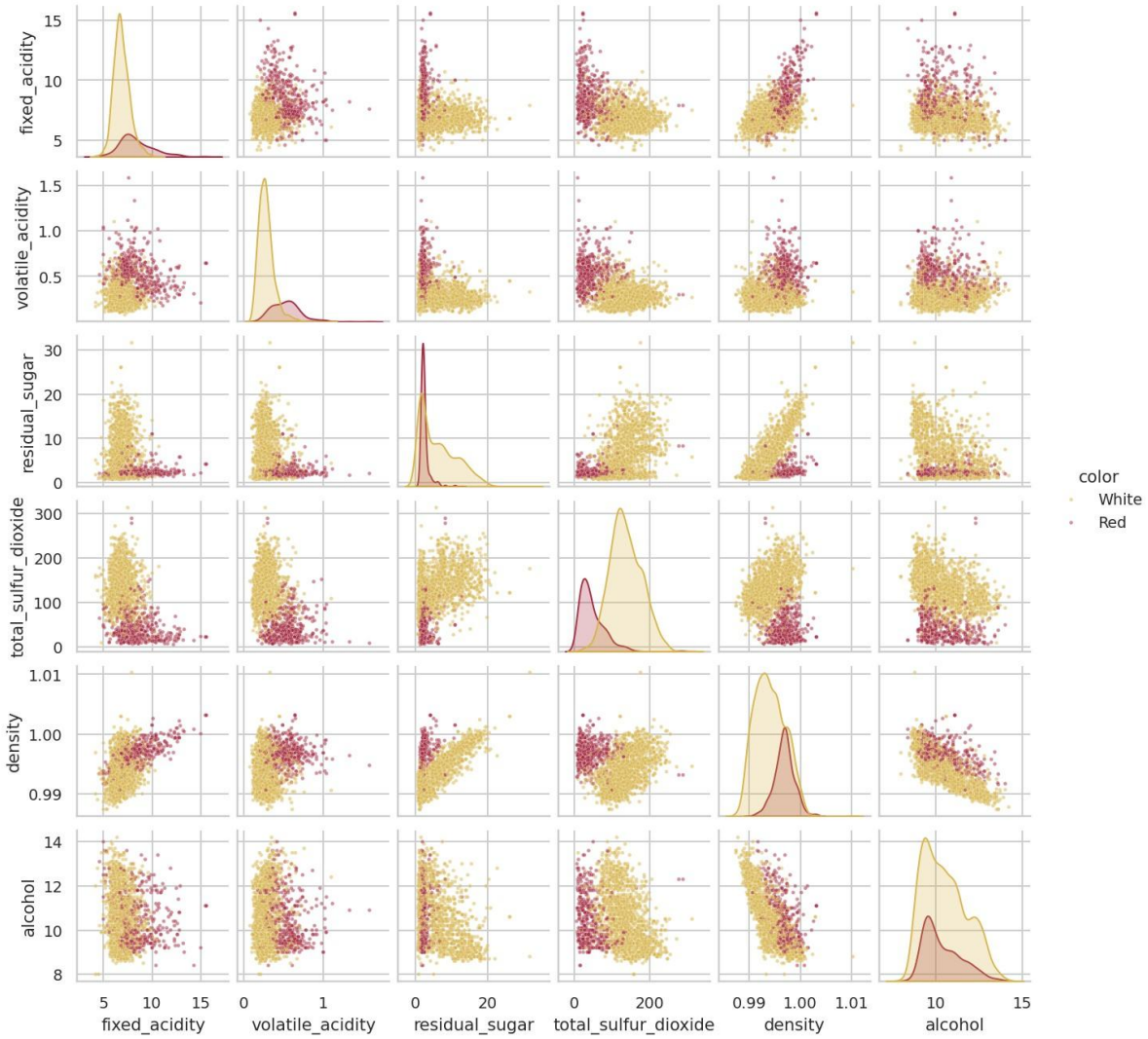


Table 2 and Figure 3. The strongest pairwise correlations are free–total sulfur dioxide (r = .72), density–alcohol (r = –.69), residual sugar–density (r = .55), and residual sugar–total sulfur dioxide (r = .50). Density is the 'hub' variable: it is pulled up by sugar and fixed acidity and down by alcohol.

Variable pair	**r**
free sulfur dioxide – total sulfur dioxide	0.72
density – alcohol	−0.69
residual sugar – density	0.55
residual sugar – total sulfur dioxide	0.49
fixed acidity – density	0.46
volatile acidity – total sulfur dioxide	−0.41
residual sugar – free sulfur dioxide	0.40
chlorides – sulphates	0.40

Figure 4. Scatterplot matrix of selected variables (random subsample, n = 2,000)



Outliers. Many univariate outliers appear by Tukey's rule (e.g., citric acid 509, volatile acidity 377, fixed acidity 357, chlorides 286), but most are in the long tails of naturally skewed variables rather than apparent errors; counts of $|z| > 3.29$ range from 1 (alcohol) to 111 (fixed acidity). Using all 11 variables, **175 wines (2.7%)** exceeded the multivariate Mahalanobis criterion ($D^2 > 31.3$, $\chi^2(11)$, $p < .001$). Extreme single values, such as residual sugar = 65.8 g/L, density = 1.039, and free sulfur dioxide = 289 mg/L (all in white wines), are plausible chemically but are influential.

Considerations before PCA/EFA. (a) **Density** is almost a linear combination of sugar, alcohol, and fixed acidity (its variance inflation factor is 15.9; all others ≤ 7.2), which risks a near-boundary (Heywood) factor solution. (b) The pooled correlation structure partly reflects the red/white mean differences, so some components or factors may be 'wine type' dimensions. (c) Skewed variables and outliers can distort correlations; I retain them for comparability with the published dataset and discuss this as a limitation. (d) The KMO value for the full set is low (Section 6.3), which calls for cautious interpretation.

5.2 MANOVA

Choice of dependent variables. Seven of the 11 variables were chosen: **fixed acidity, volatile acidity, residual sugar, chlorides, total sulfur dioxide, sulphates, and alcohol**. They span the main chemical aspects of wine (acidity, sweetness, salinity, preservation, and strength), and the exploratory analysis showed they differ visibly or plausibly between the two wine types. *Density* was excluded because it is nearly determined by sugar, alcohol, and fixed acidity (VIF 15.9), which would make the covariance matrix close to singular and add no independent information. *Free sulfur dioxide* was excluded because it is largely contained in total sulfur dioxide (r = .72).

pH and *citric acid* were excluded because acidity is already represented by fixed and volatile acidity and pH is mechanically tied to acid content. *Alcohol* was retained deliberately, as a theoretically important variable whose red/white difference is expected to be small, so the follow-up tests can show which variables truly drive the multivariate difference. The correlations among the chosen DVs range from −.41 to .50, with a correlation-matrix determinant of 0.253: the variables are moderately related, which is the situation in which a multivariate test is more informative than separate ANOVAs, and there is no multicollinearity. **Hypotheses.** H : $\boldsymbol{\mu}$ = $\boldsymbol{\mu}$ for the vector of seven DVs; H : the mean vectors differ.

0 red white 1

Assumption checks

- **Independence.** Wines were not sampled by a documented random design, and the data contain no producer or batch identifier; wines from the same producer or vintage may be more alike than wines chosen at random. The 1,177 duplicate rows could represent repeated records. This assumption can be assumed to hold only approximately, and is addressed with a duplicate-free sensitivity analysis below.
- **Multivariate normality.** Rejected: Henze–Zirkler HZ = 14.34 (red) and 9.91 (white), both $p < .001$. The χ^2 Q–Q plots (Figure 5) bend away from the reference line in the upper tail, which reflects right-skewed variables (chlorides, residual sugar, sulphates). With groups of n = 1,599 and 4,898, the multivariate central limit theorem makes the mean-vector test reasonably robust to this, and tests of normality at this sample size reject for trivial deviations; I therefore treat it as a caution, not a barrier.
- **Multivariate outliers.** Mahalanobis $D^2 > 24.3$ ($\chi^2(7)$, $p < .001$) flagged 52 red wines (3.3%) and 120 white wines (2.4%). They were retained in the main model and removed in a sensitivity analysis.
- **Homogeneity of covariance matrices.** Rejected: Box's M = 7,679.9, $\chi^2(28)$ = 7,667.2, $p < .001$; Levene's tests were also significant for every DV (all $p < .001$). The log-determinants of the covariance matrices are −4.80 (red) and −6.51 (white), so white wines (the larger group) have the smaller covariance. That combination can make F-tests liberal; this is why I use Pillai's trace (the most robust statistic) and confirm the result with sensitivity analyses. Box's M is itself highly sensitive at this sample size.
- **Relationships among DVs.** Moderate correlations (−.41 to .50) and no multicollinearity (see above), so the assumption is reasonable.

Figure 5. Chi-square Q-Q plots of Mahalanobis distances (MANOVA dependent variables)

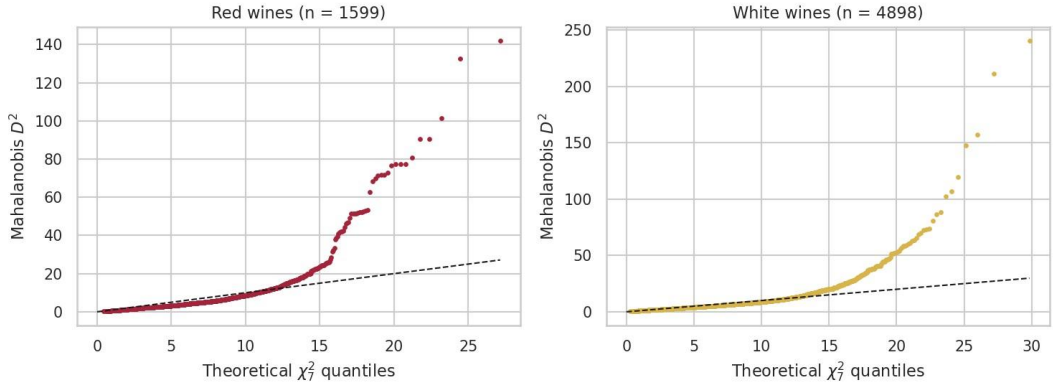


Figure 5. Chi-square Q–Q plots of Mahalanobis distances by color. Deviation above the line in the upper tail indicates heavy-tailed, right-skewed multivariate distributions.

MANOVA results

Red and white wines differ significantly on the combined profile: **Pillai's trace = 0.771, $F(7, 6489) = 3{,}118.7$, $p < .001$, multivariate partial $\eta^2 = .77$**. With only two groups, there is one discriminant dimension, so all four multivariate statistics are equivalent and give the same F (Table 3). About 77% of the variance in the best linear combination of the seven chemical variables is associated with wine color, which is a very large effect.

Table 3. Multivariate tests of the color effect (7 DVs)

Statistic	Value	F	df1	df2	p

Pillai's trace	0.771	3,118.7	7	6,489	< .001
Wilks' Λ	0.229	3,118.7	7	6,489	< .001
Hotelling-Lawley trace	3.364	3,118.7	7	6,489	< .001
Roy's greatest root	3.364	3,118.7	7	6,489	< .001

Table 4. Sensitivity of the MANOVA result to duplicates and outliers

Analysis	n	Pillai's trace	F(7, df2)	df2	p
All wines (main)	6,497	0.771	3,118.7	6,489	< .001
Duplicates removed	5,320	0.764	2,462.7	5,312	< .001
Multivariate outliers removed	6,325	0.817	4,016.1	6,317	< .001

The conclusion is unchanged: Pillai's trace is .764 without duplicates and .817 without multivariate outliers. The result is therefore not an artefact of repeated rows, extreme values, or the violated assumptions.

Follow-up analyses

Table 5. Follow-up univariate ANOVAs (Bonferroni-adjusted over 7 tests), effect sizes (η^2, Cohen's d; d > 0 means higher in red), and Welch tests

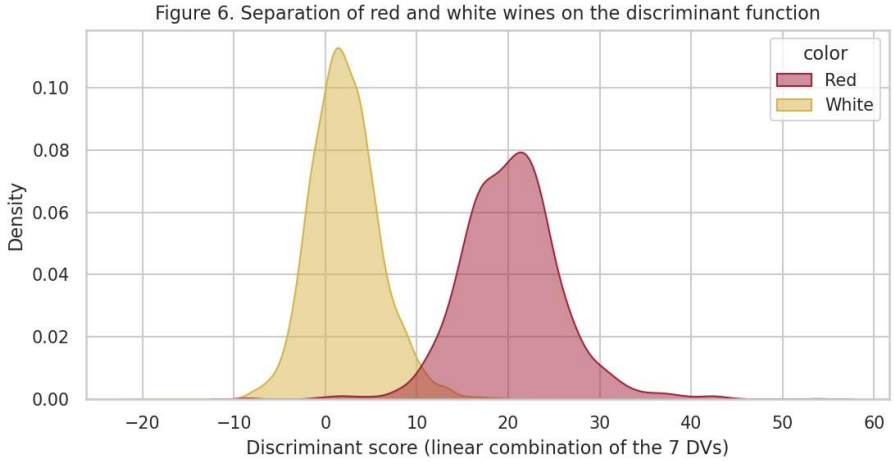
Dependent variable	M (SD) red	M (SD) white	F(1, 6495)	p (Bonf.)	η^2	d	Welch p
fixed acidity	8.32 (1.74)	6.85 (0.84)	2,016.5	< .001	.237	1.29	< .001
volatile acidity	0.528 (0.179)	0.278 (0.101)	4,829.3	< .001	.426	2.00	< .001
residual sugar	2.54 (1.41)	6.39 (5.07)	899.8	< .001	.122	−0.86	< .001
chlorides	0.087 (0.047)	0.046 (0.022)	2,315.8	< .001	.263	1.39	< .001
total sulfur dioxide	46.47 (32.90)	138.36 (42.50)	6,252.8	< .001	.491	−2.28	< .001
sulphates	0.658 (0.170)	0.490 (0.114)	2,021.7	< .001	.237	1.30	< .001
alcohol	10.42 (1.07)	10.51 (1.23)	7.1	.055	.001	−0.08	.004

Six of the seven DVs differ significantly with large effects. **Total sulfur dioxide** contributes most (η^2 = .49, d = −2.28; white wines contain roughly three times as much), followed by **volatile acidity** (η^2 = .43, d = 2.00), **chlorides** (η^2 = .26, d = 1.39), **fixed acidity** and **sulphates** (η^2 = .24 each, $d \approx 1.3$), and **residual sugar** (η^2 = .12, d = −0.86). **Alcohol** shows a negligible difference (η^2 = .001, d = −0.08); the classical ANOVA is not significant after Bonferroni correction (adjusted p = .055), whereas the Welch test is borderline (p = .004 unadjusted, about .03 after Bonferroni); either way the effect is practically trivial. Post-hoc pairwise comparisons are not applicable because color has only two levels; each univariate F-test is already the pairwise comparison.

Table 6. Discriminant function separating red from white wines (positive = red)

Variable	Standardized coefficient	Structure correlation
fixed acidity	0.30	0.55
volatile acidity	0.54	0.74
residual sugar	−0.07	−0.40
chlorides	0.19	0.58
Variable	**Standardized coefficient**	**Structure correlation**
total sulfur dioxide	−0.64	−0.80
sulphates	0.35	0.56
alcohol	−0.22	−0.04

Figure 6. Separation of red and white wines on the discriminant function



The discriminant function (squared Mahalanobis distance between the group centroids = 18.1) shows the same pattern in joint form: it is defined mainly by high total sulfur dioxide at the white pole and by high volatile acidity, chlorides, sulphates, and fixed acidity at the red pole (structure correlations of −.80, .74, .58, .56, and .55). The suppressor-like coefficients (e.g., fixed acidity .30 and alcohol −.22 despite a near-zero alcohol structure correlation) arise because the variables are correlated.

Interpretation. Red and white Vinho Verde wines are chemically different wines, not merely wines that differ on one measurement. Red wines are the more acidic, saltier (chloride-rich), and sulphate-rich wines, with higher volatile acidity (the acetic-acid fraction, associated with a vinegar note at high levels) and relatively little residual sugar. White wines contain much more sulfur dioxide (a preservative against oxidation and spoilage), possibly reflecting different winemaking practice (this dataset cannot establish the reason), and they retain more residual sugar. Alcohol strength, by contrast, does not separate the types, which shows that the multivariate difference arises from preservation chemistry, acidity, and mineral/salt content, not from fermentation strength.

5.3 PCA

Correlation structure. Bartlett's test of sphericity rejects the identity matrix ($\chi^2(55) = 37{,}094.6$, $p <$ .001) and the correlation-matrix determinant is small (.0033), so there are many substantial correlations (Figure 3). The overall KMO is, however, only **0.41**, below the conventional .50 threshold, and five variables fall below .50 (fixed acidity .28, residual sugar .29, density .31, pH .21, alcohol .27). The cause is the density hub: its strong ties to sugar, alcohol, and fixed acidity make many partial correlations large. PCA, which describes total variance and does not require a latent-variable model, is less sensitive to a low KMO than EFA, so I proceed, but I regard the components as descriptive.

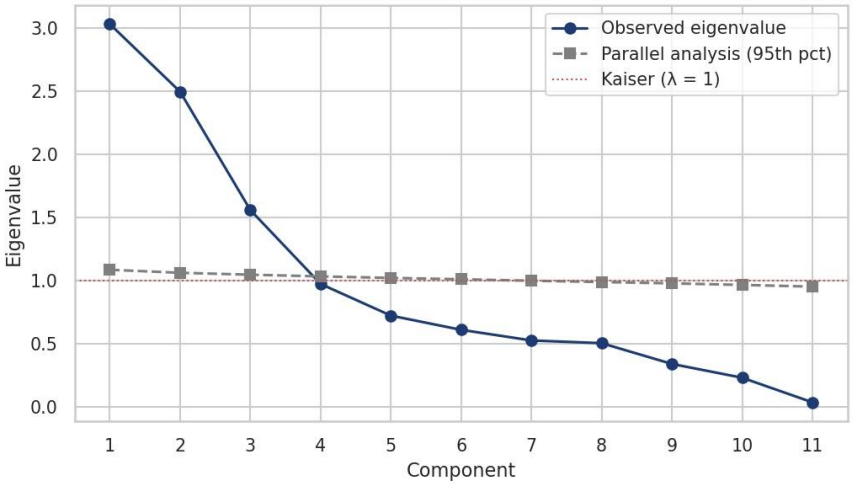
Standardization. The variables differ greatly in scale (total sulfur dioxide SD = 56.5 versus chlorides SD = 0.04, and density's SD is about 0.003). On raw or covariance-matrix PCA, the first component would simply track total and free sulfur dioxide. All variables were therefore converted to z-scores, which is equivalent to extracting components from the correlation matrix and gives each variable equal prior weight.

Table 7. Eigenvalues, explained variance, and parallel analysis (standardized variables)

Component	Eigenvalue	% variance	Cumulative %	Parallel-analysis 95th pct	Retain?
PC1	3.030	27.5	27.5	1.083	Yes
PC2	2.494	22.7	50.2	1.059	Yes
PC3	1.556	14.1	64.4	1.044	Yes
PC4	0.971	8.8	73.2	1.031	No
PC5	0.720	6.5	79.7	1.019	No
PC6	0.607	5.5	85.3	1.008	No

PC7	0.523	4.8	90.0	0.996	No
PC8	0.502	4.6	94.6	0.986	No
PC9	0.337	3.1	97.6	0.975	No
PC10	0.228	2.1	99.7	0.963	No
PC11	0.033	0.3	100.0	0.950	No

Figure 7. Scree plot with parallel analysis



Number of components. Three lines of evidence agree on **three components**. (1) Eigenvalues: PC1–PC3 are 3.03, 2.49, and 1.56; PC4 is 0.97, just below 1. (2) Scree plot: a clear elbow follows PC3, after which the eigenvalues decline gradually. (3) Parallel analysis: only the first three observed eigenvalues exceed the 95th percentile of eigenvalues from random data (1.08, 1.06, 1.04); PC4 (0.97) is below its threshold of 1.03. Because PC4 is borderline (8.8% of variance, with sulphates loading .63), the three-component solution is a parsimonious rather than clear-cut choice. The three components retain **64.4%** of the total variance (27.5%, 22.7%, and 14.1%); adding PC4 would raise this to 73.2%. Eleven variables are thereby summarized by three dimensions at the cost of about a third of the information, a moderate reduction, not a dramatic one.

Table 8. Component loadings (correlations between variables and components); |loading| ≥ .40 in bold

Variable	PC1	PC2	PC3	Communality
fixed acidity	**−0.42**	**0.53**	**0.54**	0.75
volatile acidity	**−0.66**	0.19	−0.38	0.62
citric acid	0.27	0.29	**0.74**	0.70
residual sugar	**0.60**	**0.52**	−0.20	0.68
chlorides	**−0.51**	**0.50**	−0.02	0.50
free sulfur dioxide	**0.75**	0.11	−0.17	0.60
total sulfur dioxide	**0.85**	0.14	−0.13	0.76
density	−0.08	**0.92**	−0.22	0.91
pH	−0.38	−0.25	**−0.57**	0.53
Variable	**PC1**	**PC2**	**PC3**	**Communality**
sulphates	**−0.51**	0.30	0.09	0.36
alcohol	−0.18	**−0.73**	0.33	0.68
Eigenvalue	3.03	2.49	1.56	

% variance	27.5	22.7	14.1	64.4 (total)

Figure 9. PCA loading plot (PC1 vs PC2)

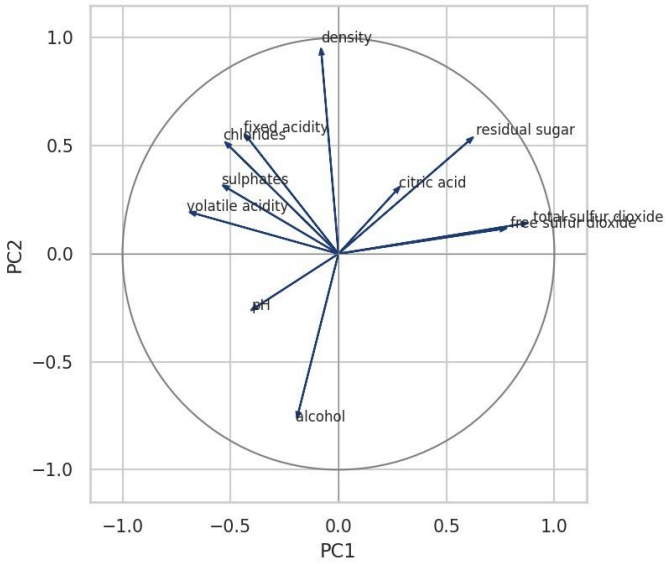
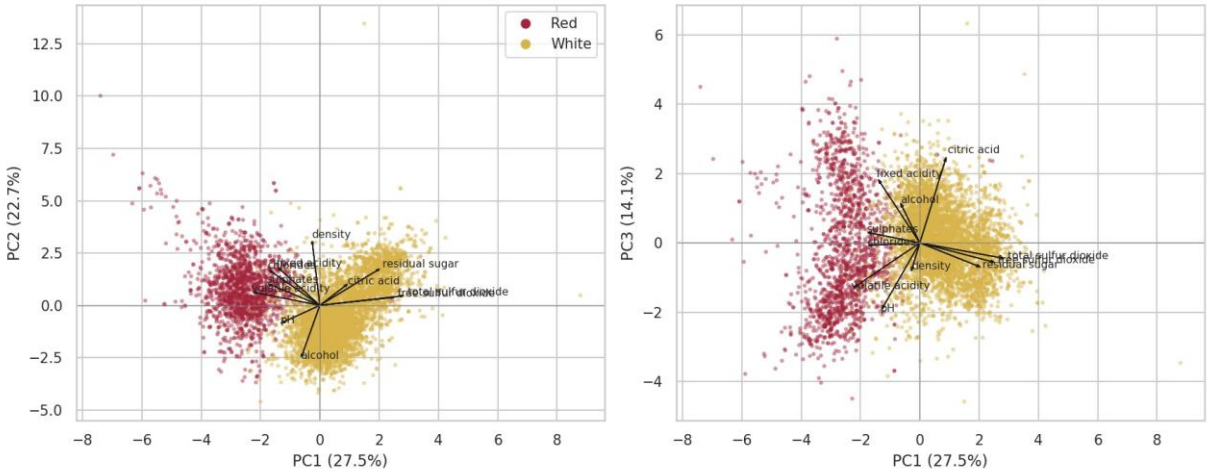


Figure 8. PCA biplots (points = wines by color; arrows = loadings, scaled)



Interpretation of the components

- **PC1 (27.5%) – 'Sulfur preservation versus acidity/salinity' (the wine-type axis).** High positive loadings: total sulfur dioxide (.85), free sulfur dioxide (.75), residual sugar (.60). High negative loadings: volatile acidity (−.66), chlorides (−.51), sulphates (−.51), fixed acidity (−.42). Total sulfur dioxide contributes most. A wine with a high PC1 score is a heavily sulfited, sweeter, less acidic wine, and a low score is an acidic, chloride- and sulphate-rich wine. This component aligns with color: red and white wines are separated by almost two standard deviations on PC1 (*d* = −1.92), and it echoes the MANOVA discriminant function.

- **PC2 (22.7%) – 'Density, sugar, and alcohol' (body and sweetness).** Density (.92), alcohol (−.73), fixed acidity (.53), residual sugar (.52), and chlorides (.50). Density contributes most. High scores indicate dense, sugary, low-alcohol wines; low scores indicate lighter, more alcoholic wines. This reproduces the physical relationship (sugar raises and alcohol lowers density) and shows a smaller red–white shift than PC1 (*d* = 0.65).

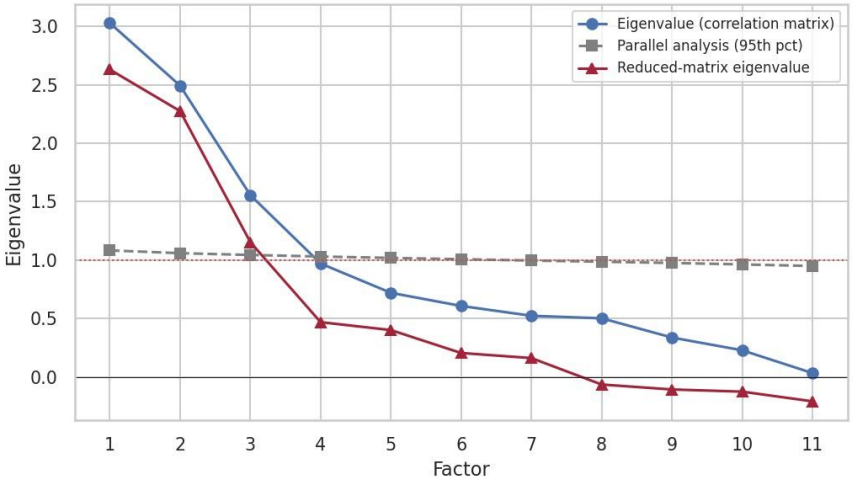
- **PC3 (14.1%) – 'Fixed acidity and freshness'.** Citric acid (.74), pH (−.57), fixed acidity (.54), volatile acidity (−.38), alcohol (.33). A high score marks wines rich in citric and fixed acid with a low pH (fresher, more tart), and a low score marks wines with higher pH and relatively more volatile acidity. It is nearly unrelated to color (d = −0.20), so it is a within-type acidity dimension.

Answer to RQ2. Three components are sufficient to summarize the 11 measurements in a useful way, retaining 64.4% of the variance. They represent (1) preservation/acidity-salinity (the red–white contrast), (2) the density–sugar–alcohol body gradient, and (3) acid freshness. Sulphates (communality .36) and chlorides (.50) are the least well represented, and sulphates has its main loading on the unretained PC4.

5.4 EFA

Factorability. The correlation matrix (Figure 3) contains many correlations above |.30| and Bartlett's test is significant ($\chi^2(55)$ = 37,094.6, p < .001), but this test is nearly always significant at N = 6,497. The overall **KMO is 0.41**, which is *unacceptable* by the usual benchmarks (≥ .50 minimum, ≥ .60 'mediocre'). Variable-level KMO values are acceptable for free sulfur dioxide (.75), chlorides (.73), and total sulfur dioxide (.72), and mediocre for citric acid and volatile acidity (.62, .61), but marginal for sulphates (.56) and poor for fixed acidity (.28), residual sugar (.29), density (.31), alcohol (.27), and pH (.21). Removing density alone raises the overall KMO to 0.66, which confirms that density is the main source of the problem. KMO is similarly low within each color (red 0.43, white 0.37). **The data are therefore only marginally suitable for factor analysis.** Because the assessment requires the 11 variables, I carry out the analysis but treat it as exploratory, and I test the stability of the result below.

Figure 11. Scree plot and parallel analysis for EFA



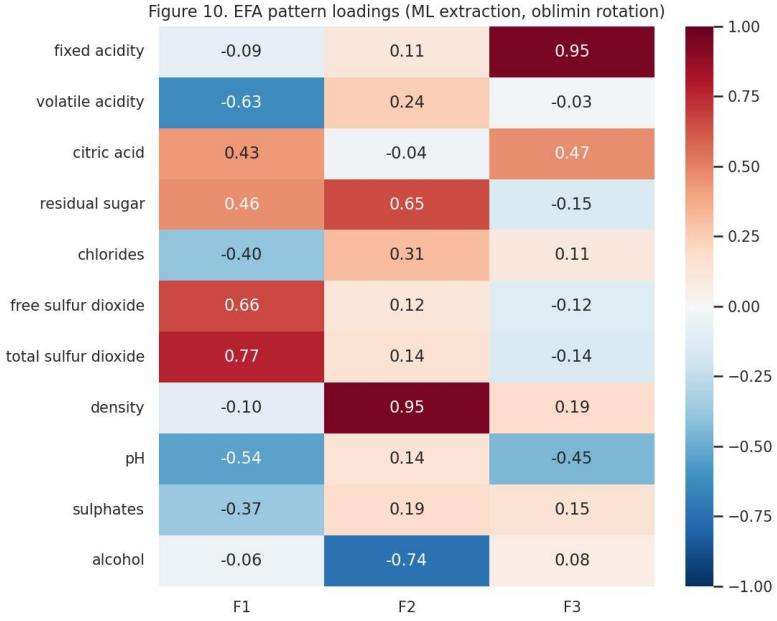
Number of factors. The scree plot has an elbow after the third eigenvalue; parallel analysis of the correlation matrix supports three (observed 3.03, 2.49, 1.56 versus random-data 1.08, 1.06, 1.04; the fourth, 0.97, is below its 1.03 threshold); and the reduced-correlation eigenvalues (2.63, 2.28, 1.15, then 0.47) have only three above 1. A parallel analysis on the reduced matrix would suggest more factors (7), but the random-data reference values are close to zero at N = 6,497, so the criterion is overly liberal and the later eigenvalues (all < 0.5) are small. Maximum-likelihood models with 1–5 factors all have significant χ^2 (expected at this N), and RMSEA falls only slowly (.30, .25, .24, .22, .19), so no small solution fits well; I chose **three factors** because it is the number supported by the scree, parallel analysis, and eigenvalue evidence, and because it gives an interpretable and parsimonious structure that parallels the PCA.

Extraction and rotation. I used **maximum-likelihood extraction** because it models the common variance only (separating it from unique variance, which PCA does not) and provides fit statistics. Its normality assumption is violated (Section 6.2), so I verified it using minres extraction, which does not require normality. **Oblimin** (oblique) rotation was used because there is no reason for chemical dimensions to be independent. The estimated factor correlations are modest (F1–F2 = .16, F1–F3 = −.22, F2–F3 = .08), so an orthogonal (varimax) rotation yields a similar solution; oblimin is retained because it does not force independence and allows the data to show it.

Table 9. EFA pattern loadings, communalities, and uniquenesses (ML extraction, oblimin rotation, 3 factors); |loading| ≥ .40 in bold

Variable	F1	F2	F3	Communality (h^2)	Uniqueness (u^2)
fixed acidity	−0.09	0.11	**0.95**	0.92	0.08
volatile acidity	**−0.63**	0.24	−0.03	0.46	0.55
citric acid	**0.43**	−0.04	**0.47**	0.41	0.59
residual sugar	**0.46**	**0.66**	−0.15	0.66	0.34
chlorides	−0.40	0.31	0.10	0.26	0.74
free sulfur dioxide	**0.66**	0.12	−0.12	0.47	0.53
total sulfur dioxide	**0.77**	0.14	−0.14	0.63	0.37
density	−0.10	**0.95**	0.18	0.95	0.05
pH	**−0.54**	0.14	**−0.45**	0.51	0.49
sulphates	−0.37	0.18	0.15	0.19	0.81
alcohol	−0.06	**−0.74**	0.08	0.56	0.45
SS loadings	2.43	2.13	1.46		
% variance	22.1	19.4	13.2	54.7 (total)	

Figure 10. EFA pattern loadings (ML extraction, oblimin rotation)



Loadings and communalities. Three factors explain 54.7% of the total variance (22.1%, 19.4%, 13.2%). *Strong loaders:* density (.95 on F2), fixed acidity (.95 on F3), total sulfur dioxide (.77 on F1), alcohol (−.74 on F2), free sulfur dioxide (.66 on F1), and residual sugar (.66 on F2). *Cross-loadings:* residual sugar (F1 .46, F2 .66), citric acid (F1 .43, F3 .47), and pH (F1 −.54, F3 −.45); volatile acidity (F1 −.63) and chlorides (F1 −.40, F2 .31) are mostly on F1 with weaker secondary loadings. *Weak variables:* sulphates (communality .19, highest loading −.37) and chlorides (.26) are poorly explained by the common factors, as is citric acid (.41), and most of their variance is unique. Density (u^2 = .05) and fixed acidity (u^2 = .08) are almost entirely explained by a single factor each, which is suspiciously close to the boundary (Heywood) case and reflects the near-redundancy of density in particular.

Interpretation of the factors

- **F1 – 'Sulfur dioxide preservation versus acidity/salinity' (wine type/style).** Total (.77) and free (.66) sulfur dioxide load positively; volatile acidity (−.63), pH (−.54), chlorides (−.40), and sulphates (−.37) load negatively, with residual sugar (.46) and citric acid (.43)

as secondary loadings. It has the same meaning as PC1, and it separates red from white wines almost completely (factor-score d = −1.90).

- **F2 – 'Density–alcohol–sugar (body)'.** Density (.95), residual sugar (.66), and alcohol (−.74). Sugar raises and alcohol lowers density, so this factor expresses the physical composition of the must/wine, i.e. its sweetness-versus-strength balance.
- **F3 – 'Fixed (non-volatile) acidity'.** Fixed acidity (.95), citric acid (.47), and pH (−.45): more acid, lower pH. This is the acid-content dimension. Red wines score higher on it (factor-score d = 0.96, driven by fixed acidity), a smaller difference than on F1 (d = −1.90); F2 shows d = 0.57.

Robustness. Minres extraction reproduces the ML factors with high congruence (Tucker's ϕ = 0.94, 0.97, 0.88) but returns a communality above 1 for density (1.03), a Heywood case, which is why ML (which bounds uniquenesses) is the reported solution. Varimax rotation gives the same three-factor interpretation with more cross-loading of fixed acidity and density. **Fit is poor** in absolute terms (ML $\chi^2(25)$ = 9,289.5, RMSEA = .24, RMSR = .075), so the three factors are a useful description, not a model that fully reproduces the correlations. Finally, I re-estimated the three-factor solution on *within-color* data (removing the red/white mean difference): the acidity factor (fixed acidity 1.00, citric acid .49, pH −.52) and a density–alcohol factor (alcohol −.89, density .64) persist, while residual sugar separates onto its own factor (.99) and the two sulfur dioxide variables no longer define a distinct factor. F2 and F3 therefore appear to represent real within-type chemistry, but much of F1 is a wine-type contrast; the within-color KMO of .38 shows that this structure is also weakly determined.

Answer to RQ3. The correlations among the physicochemical variables are explained, in broad terms, by three latent dimensions: (F1) sulfur-dioxide preservation versus acidity/salinity, which largely reflects red-versus-white style; (F2) a density–alcohol–sugar body dimension; and (F3) fixed acidity. These names are supported by the high-loading variables, but because of the low KMO, poor global fit, and the near-boundary solutions, they should be treated as exploratory.

6. Integrated Discussion

1. **MANOVA – what distinguishes red and white?** The joint chemical profile differs strongly (Pillai's trace = .77). The distinction is carried mainly by total sulfur dioxide (much higher in white), volatile acidity, chlorides, fixed acidity, and sulphates (all higher in red), and to a lesser extent residual sugar (higher in white). Alcohol does not discriminate.
2. **PCA – major dimensions.** Three components retain 64.4% of the variance: a preservation-versus-acidity/salinity axis (PC1), a density–sugar–alcohol axis (PC2), and an acid-freshness axis (PC3).
3. **EFA – latent factors.** Three oblique factors (54.7% of variance) were extracted with an interpretation that mirrors the components: sulfur-dioxide/acidity-salinity (F1), density–alcohol–sugar (F2), and fixed acidity (F3), with weak factor correlations ($|r| \leq .22$).
4. **PCA versus EFA.** Both analyses reach the same three-dimensional summary, and the same variables define the same groupings: **total and free sulfur dioxide**, **volatile acidity**, and **chlorides** (dimension 1), **density, alcohol, and residual sugar** (dimension 2), and **fixed acidity, citric acid, and pH** (dimension 3). Total sulfur dioxide, density, alcohol, and fixed acidity are the most important variables in both. The methods differ in what they model. PCA components are exact linear combinations of all the variance of the standardized variables, are by construction uncorrelated, and describe the data. EFA factors are latent causes of only the *shared* variance, allow correlated factors, and assign each variable its own unique variance. This shows in the numbers: PCA communalities are higher (e.g., sulphates .36 versus .19 in EFA, chlorides .50 versus .26), since PCA counts unique variance as component variance, and PCA retains 64.4% of the variance versus 54.7% for the factors. EFA results also depend more on the extraction and rotation choices (minres versus ML; oblimin versus varimax; the Heywood problem for density), whereas PCA is a fixed transformation of the correlation matrix.
5. **Substantive interpretation.** The chemistry of Vinho Verde wines is organized around three themes: preservation (sulfur dioxide) versus acidity and salinity, which defines the red–white difference; the density/sugar/alcohol balance, which controls body and sweetness; and fixed acidity, which controls freshness. Wine type is strongly tied to the first dimension (PC1/F1) and less so to the others. Alcohol strength does not differ between the types, although it is a key part of the density–sugar–alcohol dimension.
6. **Limitations.**

- **Weak factorability and Heywood cases.** The overall KMO is .41 (.66 without density), and the EFA fits poorly (RMSEA = .24). Density is nearly a linear function of other variables, giving boundary communalities. The factors are therefore exploratory and sensitive to modelling choices.
- **Observational data and possible dependence.** Wines were not randomly assigned or sampled by a known design; producer, vintage, and batch effects are unmeasured, and the 1,177 duplicate rows cannot be traced to unique wines. Independence is therefore uncertain, and the p-values should be read as descriptive evidence of difference, not causal claims.
- **Violated distributional assumptions and unequal groups.** Multivariate normality and equal covariance matrices were rejected, and the white group is three times larger than the red. Pillai's trace, large samples, and sensitivity analyses make the MANOVA conclusion trustworthy, but the exact p-values and effect sizes should not be over-read.
- **Outliers and skew.** About 2–3% of wines are multivariate outliers and several variables (chlorides, sulphates, residual sugar) are heavily skewed. I retained these observations (and did not transform), which may distort correlations and hence the PCA/EFA solutions.
- **Pooling red and white wines.** The combined correlation matrix mixes between-type and within-type association, so PC1/F1 partly reflects wine type rather than a single chemical process. The within-color check shows that the sulfur-dioxide dimension is the most affected.
- **Interpretation of latent factors.** Naming a factor is a judgement based on loadings; the factors are statistical constructs, not directly measured chemical entities, and are limited to the 11 variables measured (no phenolics, for instance).

Sources cited in sections 1–7

Cortez, P., Cerdeira, A., Almeida, F., Matos, T., & Reis, J. (2009). Modeling wine preferences by data mining from physicochemical properties. *Decision Support Systems, 47*(4), 547–553. https://doi.org/10.1016/j.dss.2009.05.016

Cortez, P., Cerdeira, A., Almeida, F., Matos, T., & Reis, J. (2009). *Wine Quality* [Data set]. UCI Machine Learning Repository. https://doi.org/10.24432/C56S3T (CC BY 4.0)

Methods referred to in the text (Pillai's trace, Box's M, Horn's parallel analysis, KMO, Henze–Zirkler) follow their standard formulations; the full reference list is provided in Section 8.

7. Conclusion

This study examined the physicochemical characteristics of 6,497 Portuguese Vinho Verde wines, consisting of 1,599 red wines and 4,898 white wines, using exploratory data analysis, multivariate analysis of variance (MANOVA), principal component analysis (PCA), and exploratory factor analysis (EFA).

The MANOVA results indicated a statistically significant difference between red and white wines in their combined physicochemical characteristics (Pillai's trace = 0.771, $F(7, 6489) = 3{,}118.7$, $p < .001$). Red wines generally had higher fixed acidity, volatile acidity, chlorides, and sulphates, whereas white wines had substantially higher total sulfur dioxide and residual sugar. Alcohol concentration showed little difference between the two wine types. These findings suggest that the distinction between red and white wines is associated primarily with preservation chemistry, acidity, and mineral composition rather than alcohol concentration.

PCA summarized the 11 physicochemical measurements using three principal components that collectively explained 64.4% of the total variance. These components represented sulfur dioxide preservation in contrast with acidity and salinity, the relationship between density, residual sugar, and alcohol, and a dimension associated with fixed acidity and freshness. The results demonstrate that several related chemical measurements can be represented through a smaller number of interpretable dimensions, although some information is necessarily lost.

EFA also identified three interpretable factors, accounting for 54.7% of the total variance. The factors broadly corresponded to the dimensions identified by PCA: sulfur dioxide preservation versus acidity and salinity, density–alcohol–sugar composition, and fixed acidity. However, the overall Kaiser–Meyer–Olkin (KMO) measure was only 0.41, and the factor model showed poor overall fit. Consequently, the extracted factors should be treated as exploratory descriptions rather than definitive underlying chemical structures.

Taken together, the three methods provide complementary insights into Vinho Verde wines. MANOVA establishes that the two wine types differ in their combined chemical profiles, while PCA and EFA help summarize the relationships among the measured characteristics. Nevertheless, the findings should be interpreted cautiously because of the unequal group sizes, potential dependence among observations, duplicate records, skewed variables, influential outliers, and limitations in factorability. The observational nature of the dataset also prevents causal conclusions.

Overall, the analysis demonstrates the usefulness of multivariate statistical methods for understanding complex wine chemistry. Future studies could examine additional chemical measurements, investigate the effects of producers and vintages, and validate the component and factor structures using independent samples.

8. References

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Cortez, P., Cerdeira, A., Almeida, F., Matos, T., & Reis, J. (2009). *Wine Quality* [Data set]. UCI Machine Learning Repository. https://doi.org/10.24432/C56S3T

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The pandas, NumPy, SciPy, statsmodels, scikit-learn-compatible factor_analyzer, Matplotlib, and Seaborn libraries were used in the analysis. Their use is documented in the accompanying Python script.